Directed noise can destroy ergodicity: the absorbing state in Bérard–Dembin–Marêché cannot be dropped

counterexamples-ips, entry 0003

Status: *plausible* (not yet checked by a human)

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Question. Bérard, Dembin and Marêché (arXiv:2510.11424) prove: if a finite-range monotone IPS on $\{0,1\}^{\mathbb{Z}^d}$ with $\mathbf{0}$ absorbing is ergodic, then adding *directed noise* (each site is set to 0 at rate ε, i.e. $c_0 \mapsto c_0 + \varepsilon$) gives an exponentially ergodic IPS, for every $\varepsilon > 0$. Does this remain true without the absorbing state, i.e. for every monotone ergodic L? The answer is *no*, already for plain ergodicity.

Setting. Write $0 \leftrightarrow -1$, $1 \leftrightarrow +1$ and $S_x(\eta) = \sum_{|y-x|=1}(2\eta(y)-1)$. For $\beta \geq 0$, $h \in \mathbb{R}$ let $L^{\beta,h}$ have rates

$$c_1(x,\eta) = e^{2\beta S_x(\eta)+2h} \quad (-1 \to +1), \qquad c_0(x,\eta) = 1 \quad (+1 \to -1).$$

The down-rate is constant; this is not heat-bath dynamics. The system is nearest-neighbour, monotone, with positive rates, and $\mathbf{0}$ is not absorbing.

Key identity: directed noise is a field. Adding noise towards 0 turns $c_0 = 1$ into $1+\varepsilon$, so

$$\left(L^{\beta,h}\right)_{\varepsilon} = (1+\varepsilon)\, L^{\beta,\, h-\frac{1}{2}\log(1+\varepsilon)}.$$

Proposition 1. *Let $d \geq 2$, $\beta > \beta_c(d)$, $\varepsilon > 0$, and $L := L^{\beta,h}$ with $h := \frac{1}{2}\log(1+\varepsilon)$, i.e. $c_1 = (1+\varepsilon)e^{2\beta S_x}$, $c_0 \equiv 1$. Then*

1. *L is monotone, finite range, with positive rates, and ergodic;*
2. *$L_\varepsilon = (1+\varepsilon)L^{\beta,0}$ is not ergodic: $\mu^+_{\beta,0} \neq \mu^-_{\beta,0}$ are both reversible;*
3. *$L_{\varepsilon'}$ is ergodic for every $\varepsilon' \geq 0$, $\varepsilon' \neq \varepsilon$.*

Proof. Step 1 (reversibility). In ± 1 notation the flip rate satisfies $c(x,\sigma)\, e^{\sigma_x(\beta S_x + h)} = e^{\beta S_x + h}$, independent of σ_x. So $L^{\beta,h}$ is a stochastic Ising model, and every Gibbs measure for (β,h) is reversible, hence stationary (Liggett 1985, Ch. IV).

Step 2 (sandwich). $L^{\beta,h}$ is attractive, so it is ergodic iff $\underline{\nu} = \overline{\nu}$ (Liggett 1985, Ch. III). Compare the process from $\mathbf{1}$ with the dynamics in a box Λ with spins outside frozen at $+1$. The latter dominates by the comparison theorem, and it converges to the finite-volume Gibbs measure μ^+_Λ. Letting $\Lambda \uparrow \mathbb{Z}^d$ gives $\overline{\nu} \leq \mu^+_{\beta,h}$; symmetrically $\underline{\nu} \geq \mu^-_{\beta,h}$. Hence $\mu^-_{\beta,h} \leq \underline{\nu} \leq \overline{\nu} \leq \mu^+_{\beta,h}$.

Step 3. For $h \neq 0$ the Gibbs measure is unique at every β (Lee–Yang; Lebowitz–Martin-Löf), so the sandwich gives ergodicity of L and of $L_{\varepsilon'} = (1+\varepsilon')L^{\beta,h'}$, $h' = \frac{1}{2}\log\frac{1+\varepsilon}{1+\varepsilon'} \neq 0$. For $h = 0$ and $\beta > \beta_c$ the plus and minus states differ (Peierls), and both are stationary for L_ε by Step 1. $\square$

Remark. Without Lee–Yang: the pressure is convex in h, hence differentiable off a countable set, where the Gibbs measure is unique. This gives the counterexample for all but countably many ε.

Concrete instance. $d = 2$, $\beta = \frac{1}{2} > \beta_c \approx 0.441$, $\varepsilon = 1$: flip $+1 \to -1$ at rate 1 and $-1 \to +1$ at rate $2e^{S_x}$. This is ergodic; adding rate-1 noise towards -1 makes it non-ergodic.

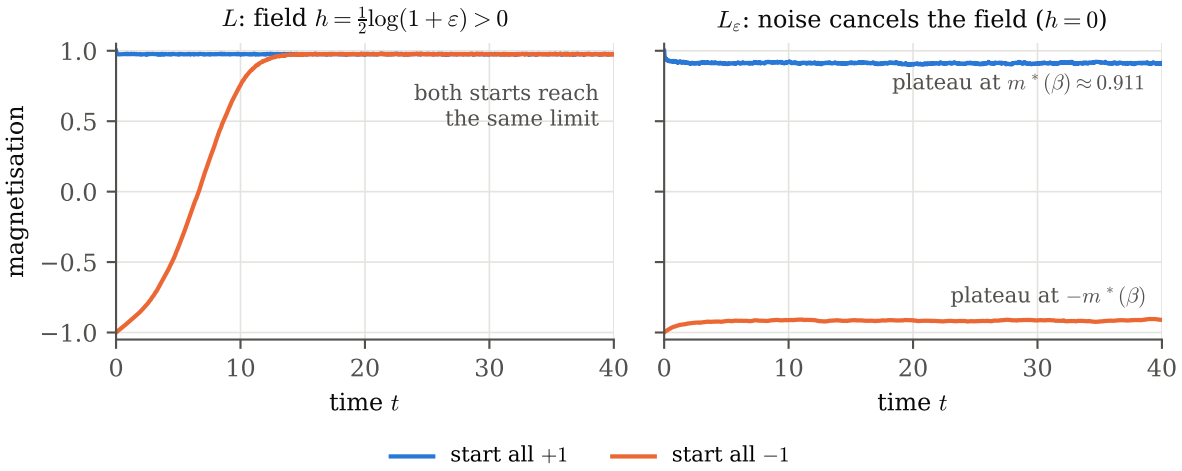


Figure 1: Simulation of the concrete instance on a 200×200 torus (`code/simulate.py`). Under L both starts reach the same limit; under L_ε they stay at the Onsager–Yang values $\pm m^*(\beta) \approx \pm 0.911$. On a finite torus these plateaux are metastable states; the figure illustrates, it does not prove.

Mechanism. Because the down-rate is constant, extra down-flips only rescale it, and a rescaled ratio c_1/c_0 is exactly a shift of the field. The noise moves the ergodic system at $h > 0$ onto the coexistence line $h = 0$. In the absorbing case the noise pushes towards the unique stationary measure $\delta_{\mathbf{0}}$ and can only help. Without an absorbing state the non-ergodic set can lie between L and L_ε, and ergodicity is not monotone in ε. With heat-bath rates the trick fails: $c_1/(c_0 + \varepsilon)$ is no longer of Ising form. (Spin-flipped version: constant rate $-1 \to +1$, noise towards $+1$.)

What this does not show. The open question of Bérard–Dembin–Marêché (Sect. 1.5) asks for *some* arbitrarily small perturbation that is exponentially ergodic. Here $L_{\varepsilon'}$ is ergodic for $\varepsilon' < \varepsilon$, so that question remains open.

Follow-up questions. See `followups.md` in the entry folder.

References

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